

Education

B.Sc. Computer Science and Engineering

The Chinese University of Hong Kong, Shenzhen

GPA: 3.77/4.00 (Dean's List, Academic Scholarship, Research Awards) Sept. 2022 – June 2026 (Expected)

Exchange Semester in Informatics (Computer Science)

Technical University of Munich

School of Computation, Information and Technology, Informatics Department

April 2025 – August 2025

Publications

Mingxuan Wang, Satoshi Nakamura[†]. “TokenChain: A Discrete Speech Chain via Semantic Token Modeling”. *IEEE Intl. Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026 📄 Under review 📄 [arXiv](#)

Dingjie Song, Sicheng Lai, **Mingxuan Wang**, Shunian Chen, Lichao Sun, and Benyou Wang[†]. “Both Text and Images Leaked! A Systematic Analysis of Data Contamination in Multimodal LLM”. *Conference on Empirical Methods in Natural Language Processing (EMNLP) Findings*, 2025 📄 ICML DIG-BUGS Oral 📄 [arXiv](#)

Liumeng Xue, Chaoren Wang, **Mingxuan Wang**, Xueyao Zhang, Jun Han[†], and Zhizheng Wu[†]. “SingVisio: Visual analytics of diffusion model for singing voice conversion”. *Computers & Graphics*, 2024 🔗 [DOI](#)

Xueyao Zhang, Liumeng Xue, ..., **Mingxuan Wang**, ..., Zhizheng Wu[†]. “Amphion: an Open-Source Audio, Music, and Speech Generation Toolkit”. *IEEE Spoken Language Technology Workshop (SLT)*, 2024 🔗 [DOI](#)

Research Experience

TokenChain: A Discrete Speech Chain via Semantic Token Modeling

November 2024 – Present

Lead Researcher, CUHK-Shenzhen Spoken Language Comms Lab

Supervisor: [Prof. Satoshi Nakamura](#)

- Pioneered the research question and led the end-to-end project: a fully discretized semantic token ASR–TTS machine speech chain with straight-through (STE) / Soft-Gumbel feedback and dynamic loss scheduling.
- Demonstrated effectiveness with discrete-token chaining: in-domain beat baseline by 26 epochs with 5–13% lower equal-epoch error; out-of-domain cut ASR WER 56% and T2S WER 31% with minimal forgetting.
- Led an independent, first-author effort: engineered a unified preprocess-training-evaluation framework, ablated hyperparameters and produced figures, and submitted the manuscript to IEEE ICASSP 2026. 📄 [arXiv](#)

Semantic-Syntactic Aware Mixture-of-Experts (MoE) LLM

March 2025 – August 2025

Visiting Undergraduate Researcher, TUM Research Group Social Computing

Supervisor: [Prof. Georg Groh](#)

- Architected innovative MoE LLM with semantic-syntactic separation mechanism, designed label-aware and adaptive routers with load-balancing and mask, achieving expert-set specialisation with strong results.
- Conceived knowledge distillation and compression methods (router-side SLAR; expert-side MoKE) for the new model, delivering 1.2x faster inference and 35% lower memory with only 0.41% performance loss.
- Built the full training and evaluation stack with metrics and interpretability tools (routing heatmaps and word banks), ran ablations and authored the poster and research report documenting findings. 📄 [Report](#) 📄 [Poster](#)

Performance Evaluation of Multimodal LLM Benchmark

October 2024 – November 2024

Independent Researcher, CUHK-Shenzhen School of Data Science (SDS)

Supervisor: [Prof. Benyou Wang](#)

- Proposed novel contamination-transfer ablations investigating unimodal and its derived multimodal models.
- Improved key correct-answer matching algorithm for leakage detection, boosting selection accuracy by 20%.
- Optimized contamination detection by inhibiting random guessing via uncertainty clause, analyzed back-translation quality through semantic retention and lexical overlap with SBERT, BLEU and their correlation.

- Accelerated Amphion & SingVisio’s TTS development by optimizing code for critical evaluation milestones.
- Implemented Mamba SSM text-to-semantic models attaining higher speaker similarity than Llama baseline.
- Developed expertise in two-stage discrete tokenized TTS paradigm, gaining insights into masked-generative text-to-semantic modeling, RVQ-based speech tokenisation, and semantic-to-acoustic synthesis workflows.

Internship

Data Analytics & Cloud Intern

The Hong Kong Research Institute of Textiles and Apparel (HKRITA)

August 2023

Hong Kong SAR, China

- Managed the Enterprise Resource Planning (ERP) system, ensuring operational workflows of 20+ projects.
- Led and successfully completed a feasibility study for the company’s adoption of a cloud-based service.
- Collaborated with Microsoft and key partners to formulate a plan for seamless migration of service, gaining insights into cloud-based system platforms and strategic data analytics architecture planning.

Projects

SingVisio: Visual Analytics of Diffusion Model for Singing Voice Conversion

Key Contributor, CUHK-Shenzhen Human Language Technology Lab

January - March 2024

 [Application Website](#)

- Calculated objective criteria of audio data to visualize changes in the diffusion model training process.
- Designed and analyzed 20+ user surveys from industry experts for system evaluation and further revisions.
- Co-authored research paper and produced video demo on SingVisio, effectively promoting project findings.

Amphion: An Open-Source Audio, Music and Speech Generation Toolkit

Key Contributor, CUHK-Shenzhen Human Language Technology Lab

October - December 2023

 [Github](#) (9.5k Stars)

- Created a novel celebrity speech dataset, supporting the lab’s future speech generation research.
- Improved an evaluation pipeline for text-to-speech models, boosting the benchmark efficiency by 30%.
- Coordinated the innovation and enhancement of UniCATS, a large-dataset text-to-speech model, enhancing model versatility by realizing multi-speaker functionality and on-the-fly audio resampling. [Code](#)

Honors

Undergraduate Research Awards (SLP GenAI)	Honored for key research contributions	2024 – 2025
Dean’s List	Recognition for academic excellence	2023 – 2025
Academic Performance Scholarship (Top 3%)	Outstanding academic merit award	2023

Qualifications

International English Language Testing System (IELTS)	8.5/9.0, CEFR Level C2	2025
Japanese Language Proficiency Test (JLPT) Level N1	Equivalent to CEFR Level C1	2023

Core Competencies

Models & Paradigms	Zero-shot AR&NAR TTS, ASR, MoE, MLLM, discrete codec, RVQ, Mamba(SSM)
DL Frameworks	Hands-on with PyTorch, mixed precision, distributed data-parallel training
MLOps & Tooling	Git, Docker, Kubernetes, SLURM, Conda, Weight & Biases, Huggingface
Data Science	NumPy, Pandas, Matplotlib, Seaborn for rapid data analysis & visualisation
Cloud Architectures	Experienced in development workflows on public, private, and on-premise clouds